Feature Selection and Dimensionality Reduction

Machine Learning, Homework 4

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Notebook (`breast_cancer_features.ipynb`). The data are the Wisconsin diagnostic breast-cancer data: 569 samples, 30 features, malignant or benign.

- *SelectKBest* (F-test) kept the size and concavity features (radius, perimeter, area, concavity, concave points; mean and worst values).
- *RFE* with logistic regression kept a different set, including texture, smoothness, compactness and symmetry terms. RFE ranks features by their weight in a multivariate model, while the F-test scores each feature alone, so correlated size features are not all kept by RFE.
- A *random forest* gave feature importances; Figure 1 shows the top ten.
- *PCA* to two components explains 99.8% of the variance. This is because the features were not standardised, so the area features (values in the hundreds to thousands) dominate. *t-SNE* separates the classes more clearly than this PCA projection.

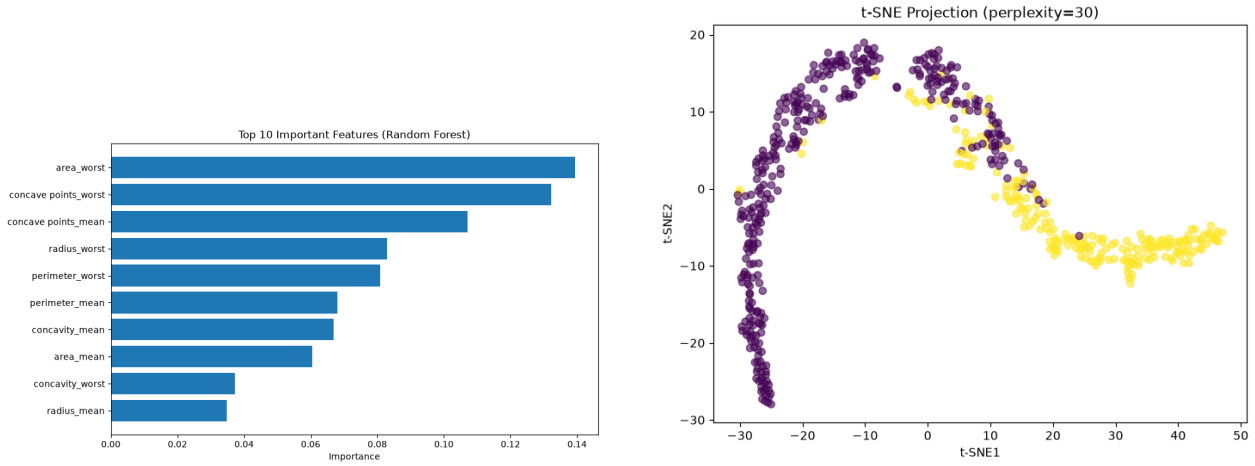


Figure 1: Random-forest importances (left) and t-SNE embedding coloured by diagnosis (right).

Written part.

PCA of three points. For $(1,1),(0,0),(-1,-1)$ the mean is 0 and the first principal axis is $(1,1)/\sqrt{2}$. The projections are $\sqrt{2}, 0, -\sqrt{2}$, with variance $4/3$ (dividing by n). The points lie on that axis, so the reconstruction error is zero.

PCA vs. Fisher LDA. PCA keeps the direction of largest spread and ignores labels. LDA picks the direction that best separates the two classes, so it can differ from PCA.

Forward and backward search. I explained greedy forward selection and wrote the backward version: start from all K features and, at each step, remove the feature whose removal increases the validation error least.

Knapsack. Choosing features under a budget is like a knapsack problem and resembles wrapper methods. Dynamic programming finds the global optimum for small integer instances, but the problem is NP-hard, so large instances use greedy heuristics without that guarantee.

PCA and kernel PCA of four points in $\mathbb{R}^4$ projected to one dimension; for kernel PCA with $k(x,y) = (x^\top y)^2$ I eigen-decomposed the 4×4 Gram matrix.